



TWITTER SENTIMENTS ON GOOGLE AND APPLE PRODUCTS

BY: STEPHEN JILANI

BUSINESS OVERVIEW

- ▶ A Kenyan company is in the predicament of classifying twitter reviews on Google and Apple products and would like to build a model that can assist in sentiment classification.

BUSINESS OBJECTIVES

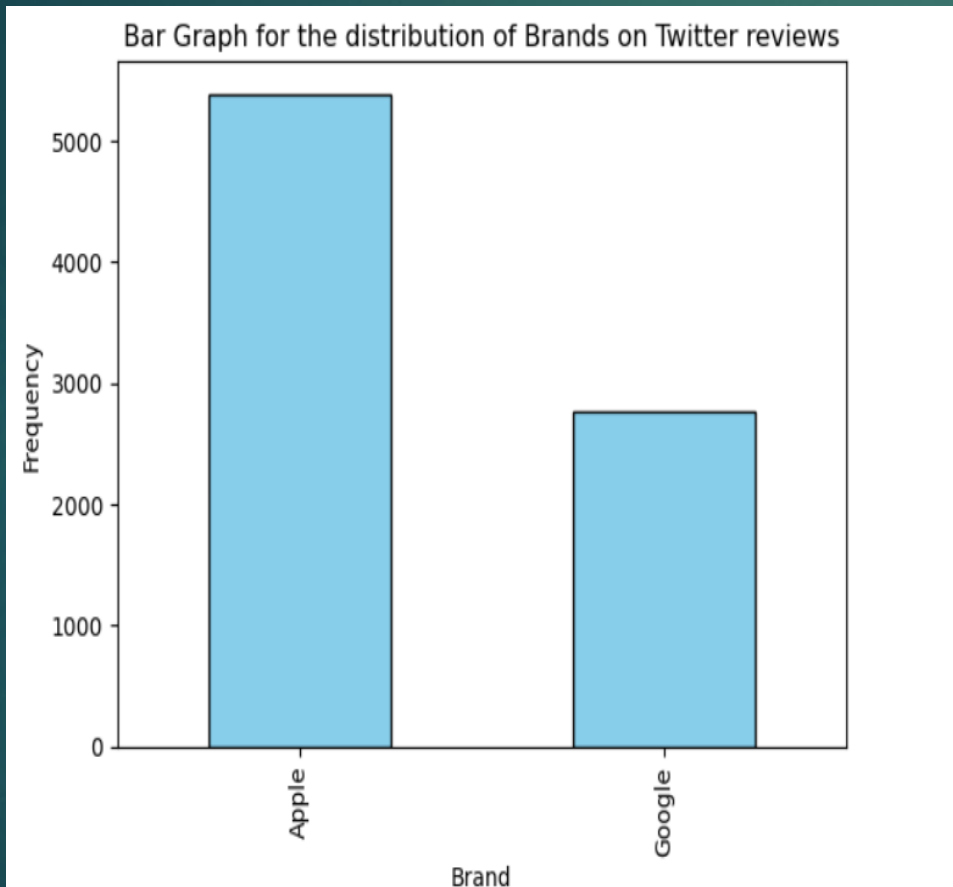
- ▶ Build a classification model that helps in the classification of twitter reviews.
- ▶ Compare multiple model performance based on F1 score, precision and recall.
- ▶ To provide insights on customer sentiment distribution.

DATA UNDERSTANDING



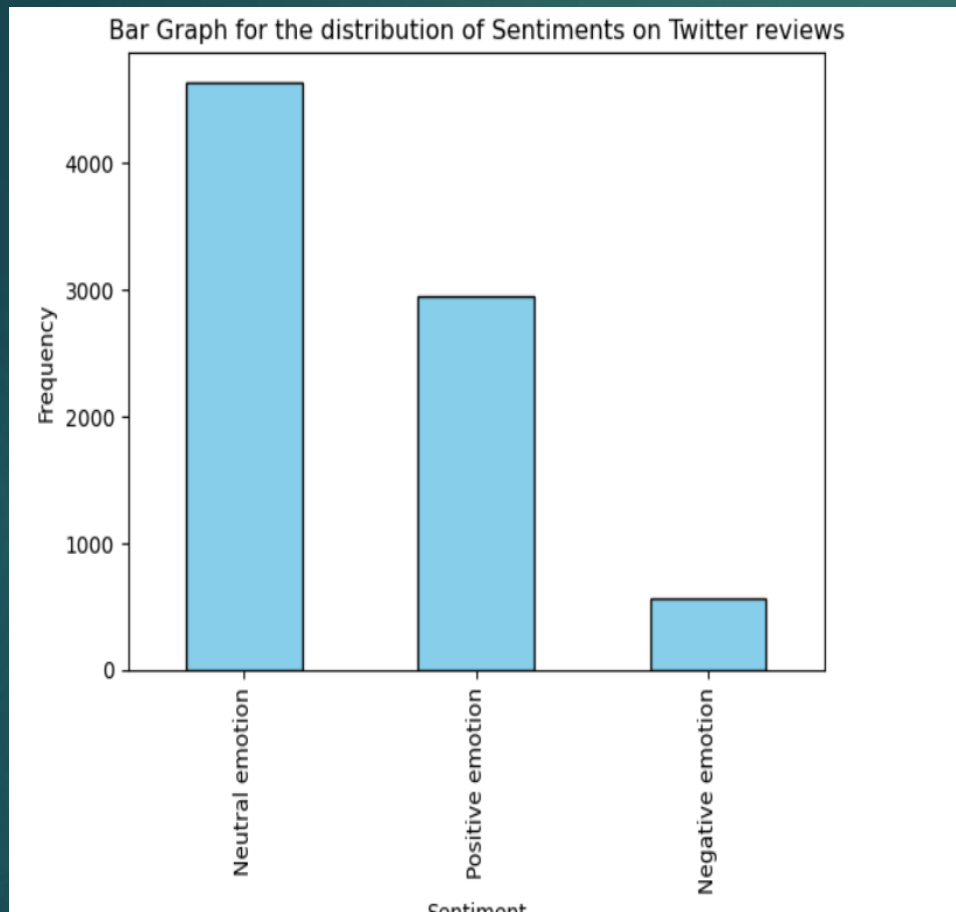
- ▶ The data was obtained from CrowdFlower via dataworld and is composed of sentiments in over 9,000 tweets in relation to Apple and Google products.
- ▶ The data contains 3 columns i.e tweet text, Brand and Sentiment classification

EXPLORATORY DATA ANALYSIS



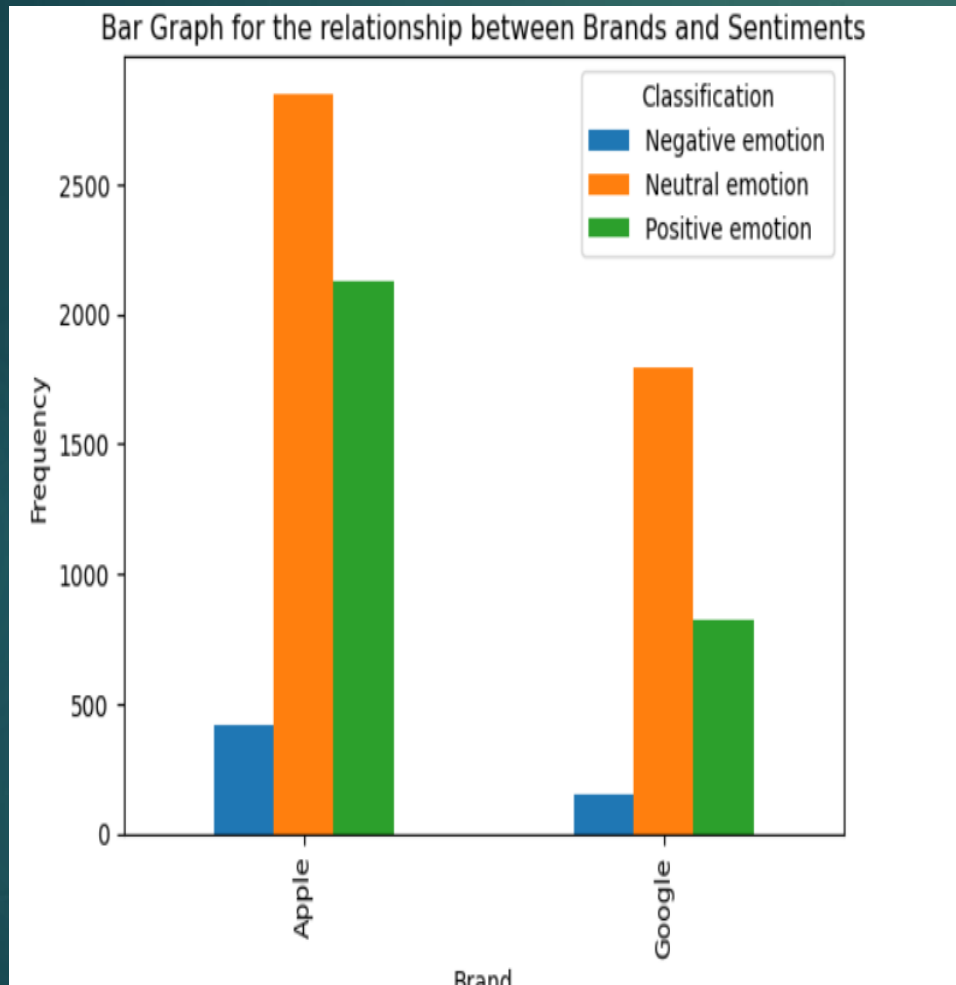
- The graph shows that most of the reviews are directed toward Apple compared to Google

EXPLORATORY DATA ANALYSIS....contd



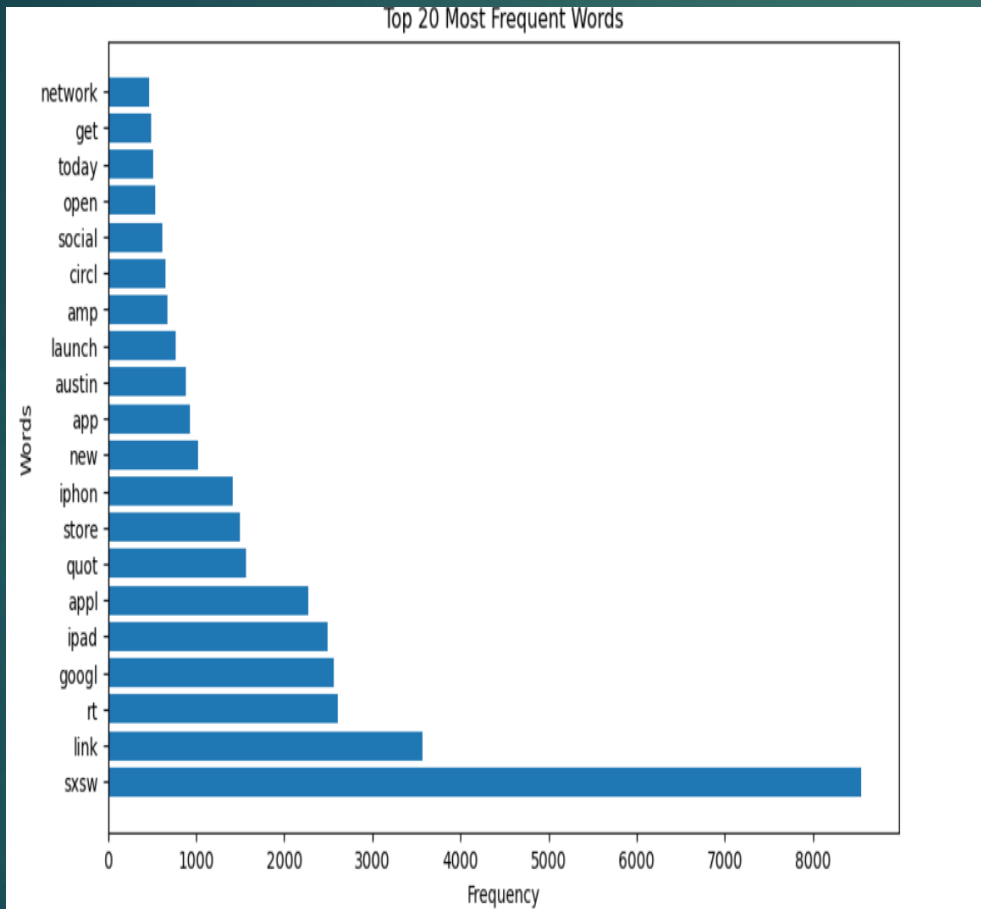
- The graph shows that most of the reviews have been classified as neutral indicating indifference in the products.

EXPLORATORY DATA ANALYSIS....contd



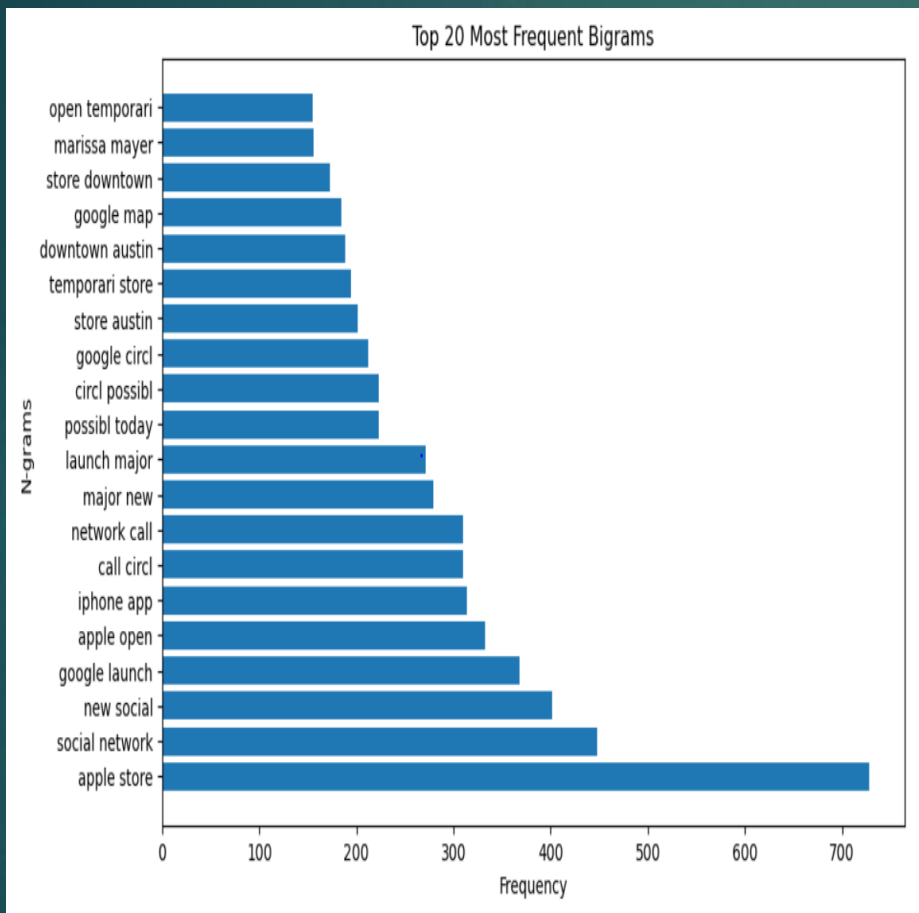
► The graph shows the distribution of reviews between Apple and Google together with their respective sentiments.

EXPLORATORY DATA ANALYSIS....contd



▶ The graph shows the the most frequent words in the dataset which may be the biggest drivers of sentiment classification.

EXPLORATORY DATA ANALYSIS....contd

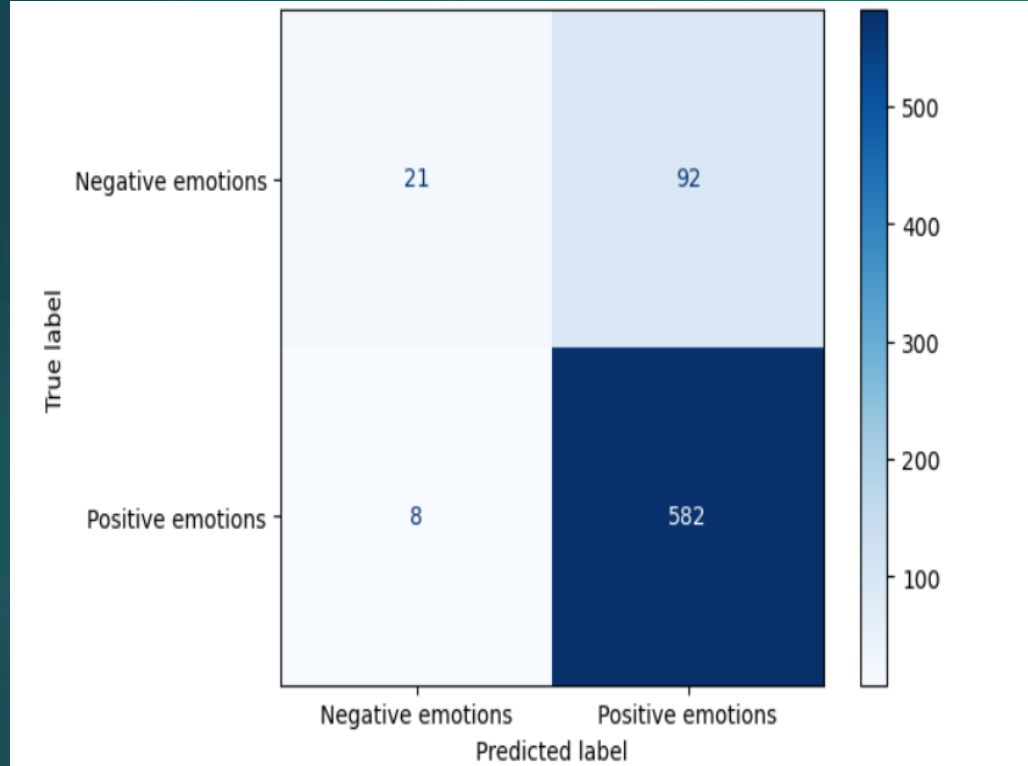


► This is a visualization of the most frequent phrases in the dataset.

DATA MODELLING AND EVALUATION

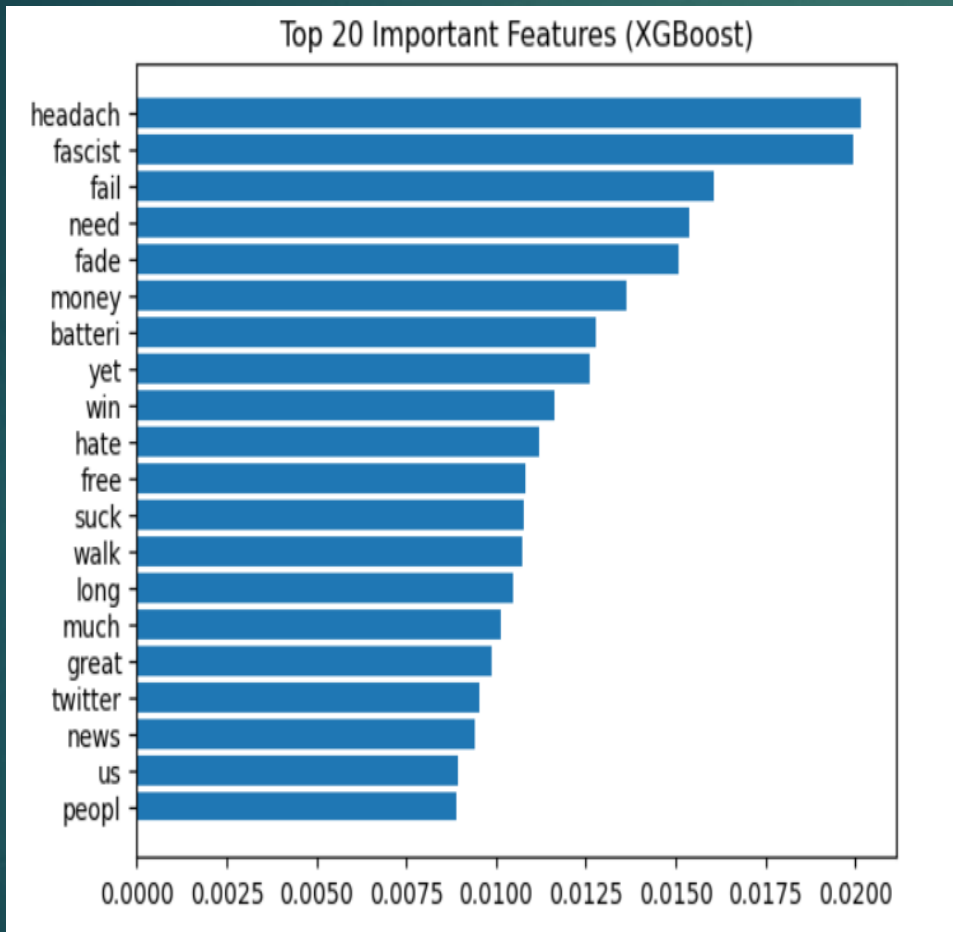
- ▶ 3 binary classification models were explored i.e Logistic Regression, RandomForest and XGBoost. Both RandomForest and XG Boost were fine tuned.
- ▶ 1 multiclass classification model was explored i.e XGBoost and it was fine tuned.
- ▶ The best models have been discussed below

DATA MODELLING AND EVALUATION....contd



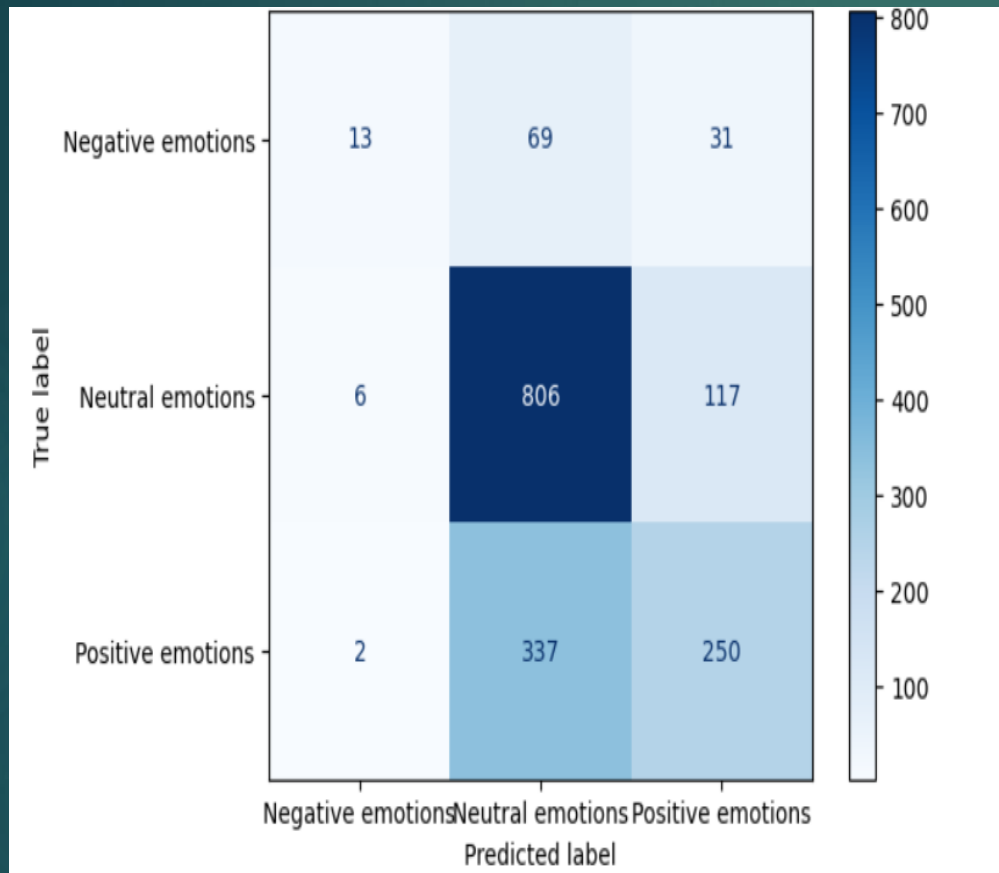
▶ The confusion matrix is for the best binary(tuned XG Boost) classification model.

DATA MODELLING AND EVALUATION....contd



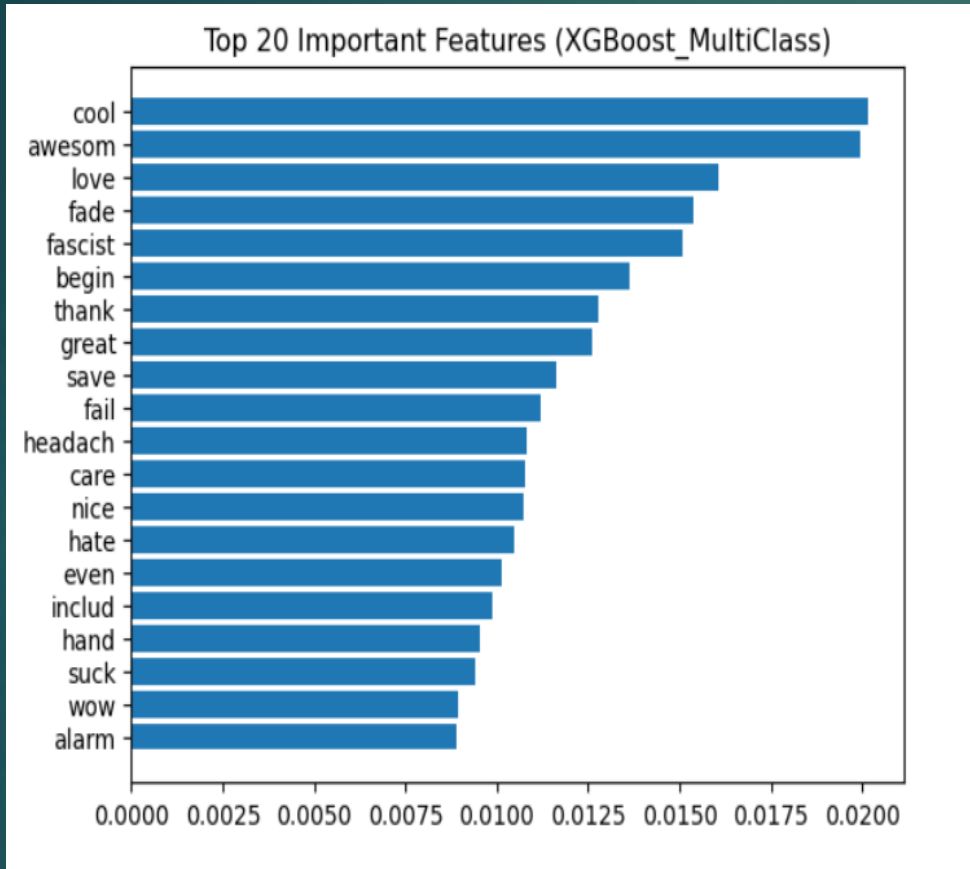
► The graph is for the 20 most important features in the binary classification problem.

DATA MODELLING AND EVALUATION....contd



▶ The confusion matrix is for the best multiclass model (tuned XGBoost) classification model.

DATA MODELLING AND EVALUATION....contd



► The graph shows the most important features in the multiclass classification model.

BUSINESS RECOMMENDATIONS

- ▶ 1. The stakeholders should implement the best models in the classification of review sentiments.
- ▶ 2. The stakeholders need to analyse top features that drive negative emotions and identify what exactly lead to such sentiments.
- ▶ 3. The stakeholders should focus more on negative sentiments and route them to customer support and product teams.

LIMITATIONS

- ▶ - Class imbalance may affect the performance of the model
- ▶ - The data was social media noisy and that may affect the performance of the model
- ▶ The data may be from a specific region in the world and may not be fit for the stakeholder's region.



THANK YOU



QUESTIONS